Agri-CNX-Edge: A Unified DINOv3-ConvNeXt Framework with LoRA Adaptation for Multi-Task Apple Leaf Disease, Pest, Fruit and Yield Analysis on Edge

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Abstract

Field deployment of agricultural vision usually means one model per task: disease, pest, fruit grade, and counting each run separately, repeating feature extraction on hardware that cannot afford it. Agri-CNX-Edge unifies leaf disease classification, pest recognition, fruit classification, and apple detection/counting in a single DINOv3-ConvNeXt-Tiny student – the central focus of this work – with LoRA adapters, FPN-Lite neck, YOLO-style detection head, and ROI-Align per-apple fruit head, supervised by ConvNeXt-Small and YOLO26s teachers with adaptive homoscedastic task weighting. On the held-out test set the proposed configuration scores 100.00% leaf, 99.56% pest, and 95.69% fruit accuracy, with counting MAE 3.53, RMSE 6.13, Pearson r=0.85, and R^2=0.57 on 400 yield images. The no-distillation ablation matches or exceeds it on all five metrics with fruit 99.14% and MAE 3.08, and the frozen backbone ties it: under saturated teachers and ceiling tasks, distillation does not help, while static weighting collapses fruit accuracy to 70.26%, validating adaptive weighting as essential. The 120.7 MB ONNX model runs fully offline on a Samsung A17 5G at 5.6–5.7 s per inference with provider-identical numerics. All runs use a single seed, and run-to-run variation is estimated at $\sim$3 percentage points on fruit.

Keywords: Agricultural artificial intelligence, Multi-task learning,

Knowledge distillation, Edge AI, Apple disease detection, Pest recognition, Fruit classification, Fruit localization, Fruit counting, DINOv3-ConvNeXt, YOLO, ROI Align

1. Introduction

Agriculture plays a critical role in food security, economic development, and sustainable resource management. Crop diseases, pest infestations, and inaccurate assessment of fruit production significantly affect crop quality and productivity, and computer vision offers a path to automate disease recognition, pest identification, fruit classification, fruit localization, and fruit counting.

Deep learning has become an important approach for agricultural image analysis because it learns discriminative visual representations directly from images. Convolutional networks, transfer learning, object detection, and self-supervised representation learning have demonstrated strong performance across agricultural applications.

However, most deployed systems are single-task: a disease-classification model, a pest-detection model, and a fruit-counting model each extract their own features independently. On resource-constrained hardware this means repeated computation, higher memory, and a deployment story that does not scale to the field. Multi-task learning promises an alternative – several related tasks sharing one visual representation – but agricultural tasks disagree in output structure and loss character: leaf and pest recognition are image-level classification while fruit counting is inherently spatial. Two further assumptions in the literature deserve testing rather than trust: that teacher distillation necessarily helps the student, and that on-device deployment follows from a small checkpoint. This work tests both claims with measurements.

Agri-CNX-Edge addresses the problem with a shared DINOv3-ConvNeXt-Tiny visual backbone as its main focus with LoRA adaptation, hierarchical feature routing, an FPN-Lite neck, lightweight task-specific heads, a decoupled anchor-free YOLO detection head with ROI-Align-based per-apple classification, heterogeneous knowledge distillation, and adaptive homoscedastic task weighting – all evaluated on-device on a budget smartphone.

The principal contributions of this work are:

- A unified multi-task architecture for leaf disease classification, pest recognition, fruit classification and fruit localization with counting that runs fully offline on device.
- Hierarchical feature routing in which fine-grained features serve leaf classification, mid-scale features serve pest recognition and fused multi-scale features serve apple detection, with per-apple fruit classification from ROI-Align crops as defined in Section 4.
- A heterogeneous distillation strategy that combines categorical distillation for classification with box-level distillation for detection, reported with the measured finding that it does not improve the student under saturated teachers.
- A teacher pipeline with three classification seeds and eight augmentation views per seed, complemented by a YOLO26s detection teacher for bounding-box supervision.
- Adaptive uncertainty weighting for heterogeneous objectives, validated as essential because static weighting collapses fruit test accuracy to 70.26%.
- A structured ablation of distillation, weighting, adaptation and warm-up with measured deployment evidence on a Samsung A17 5G with identical numerics across providers.

The remainder of the paper follows this contribution order as a single narrative: Section 2 and Table 1 show why prior work leaves the gap open, Section 3 formalises the gap, Section 4 with Figs. 1–3 and Eqs. 1–23 builds the Agri-CNX-Edge answer centred on DINOv3, Sections 5–6 fix data and measurement, Section 7 with Tables 3–13 tests each claim, and Sections 12–12 synthesise what survived measurement.

2. Related Work

This section reviews only work directly relevant to a unified multi-task model that runs offline on a phone. It proceeds from agricultural multi-task systems to offline mobile systems and finally to the representations and training mechanisms that make unification possible. Table 1 summarises the comparison and is discussed at the end of the section.

2.1. Multi-Task Learning in Agriculture

Single-task apple models remain common, such as real-time apple leaf systems [1] and orchard detection and counting pipelines [2]. The literature after

2020 moves beyond them toward joint models. Luan et al. proposed ALMDR, a multi-task apple leaf model that jointly performs multi-label classification, detection, segmentation and severity estimation with a group feature pyramid, reporting 93.74% F1, 51.32% detection mAP and severity R^2 0.9757 at 6.25 frames per second [3]. Kang et al. proposed a lightweight tomato model that jointly performs leaf disease detection and fruit counting [4], the closest precedent to combining classification with counting. PMJDM jointly performs plant species classification and disease detection from a shared ConvNeXt backbone with dynamic weight adjustment, reporting 71.84% precision and 61.83% mAP50 at 113 frames per second with 49.1M parameters [5]. PlantSDS-Net jointly performs species, disease and severity classification in one model with task attention, reporting 99.89%, 99.58% and 92.18% on the three tasks [6]. Beyond classification, CMTL unifies cotton boll detection, pest damage segmentation and growth stage classification in one transformer with cross-level encoding, reporting 0.913 mAP, 0.832 IoU and 0.936 accuracy with Jetson Xavier NX edge trials [7]. For joint disease and pest recognition, a DenseNet-121 and ResNet-50 fusion reports 99.13% disease and 98.67% pest accuracy [8], while DAM-Net uses a discriminative attention network with ResNet-50 and adaptive weighted loss to reach 99.7% on D0 and 74.1% on IP102 [9]. Pest-YOLO jointly addresses dense tiny pest detection and counting with focal loss and a confluence strategy, reporting 69.59% mAP and counting RMSE 0.44 [10]. Together these studies prove that sharing features across agricultural tasks is feasible, but none combines leaf disease, pest, fruit grading and spatial counting in one DINOv3 edge model with tested task weighting.

2.2. Mobile and Offline Edge AI in Agriculture

Field deployment requires offline inference on low-end phones, a constraint addressed explicitly only after 2020. PlanteSaine integrates EfficientNetB3 with an artificial neural network classifier into an Android app with TensorFlow Lite and offline operation for maize, tomato and onion farmers in Burkina Faso [11]. At model level, an ultra-lightweight network reports competitive plant disease and pest accuracy with fewer than 150,000 parameters [12], and a MobileNetV3-small edge model reports 99.50% accuracy after quantization to 0.93M parameters in ONNX format at 100 frames per second on CPU [13]. These systems establish that offline mobile diagnosis is achievable, but they remain single-task classifiers without unified counting, without heterogeneous distillation and without measured multi-task ablation.

2.3. Foundation Representations, Distillation and Task Balancing

DINOv2-FCS builds fruit leaf classification and severity prediction on a DINOv2-B backbone with class-patch fusion and explicit multi-scale fusion, reporting 99.67% classification accuracy, 90.29% mIoU and 95.68% severity accuracy, with mobile deployment listed as future work [14]. EConv-ViT fuses ConvNeXt and Vision Transformer with channel attention for apple leaf disease, reporting 99.2% in lab and 79.3% in natural conditions [15]. DINOv3 scales self-supervision to 7B parameters and 1.7B images and distils to ConvNeXt Tiny, Small, Base and Large variants for resource-constrained deployment [16]. This work uses the Tiny variant as the shared representation. For training, distillation is also shown in lightweight plant models [17], and task balancing follows homoscedastic uncertainty weighting [18] as the lightest alternative to gradient normalization [19] and conflict-averse updates [20]. On-device benchmarking follows MLPerf Mobile [21] and MDLBench practice [22] with a uniform app, model, images and code per device. This paper follows the same discipline while reporting the reduced sample tail honestly.

2.4. Orchard-Scale Detection and Counting

Counting is the spatial half of the gap, and orchard literature treats it as a detection problem first and an arithmetic problem second. Jin et al. enhance YOLOv8n with re-parameterised blocks, a generalised feature pyramid and a new IoU loss for robotic harvesting, reaching 81.68% mAP50 with counting accuracy 69.39% and millimetre-level localisation under varied lighting and occlusion [23]. This study proves that modern YOLO detectors survive orchard clutter, yet it stops at detection plus counting without leaf, pest or grading tasks. For temporal context, Hani et al. comparatively benchmark detection against tracking-based yield mapping methods in apple orchards [24]. This work establishes that frame-to-frame association matters as much as per-frame precision, a lesson this work inherits through decoded-box counting even though it counts from single images rather than video. Taken together, the detection literature offers mature spatial machinery with measured MAE and RMSE discipline that this work adopts, but no study routes detection features onward into per-apple grading inside one shared backbone.

2.5. Datasets, Single-Task Foundations and Field Realism

No multi-task claim is credible without naming the data it stands on. The PlantVillage repository of over 54,000 controlled leaf images across 38 classes underpins modern classification work [25]. Early reviews map the

territory: Ferentinos surveys deep models for disease detection and diagnosis [26], Dhaka et al. survey convolutional networks for leaf disease prediction [27], and Liu and Wang review deep learning for joint plant disease and pest detection [28]. Recent syntheses raise the bar further: Shoaib et al. review deep learning for plant disease and pest detection and report classification accuracies above 95% with detection precision above 90% in controlled settings [29], while Banerjee et al. taxonomise hyperspectral, visualisation, modified deep and transformer methods and show hierarchical vision transformers above 99.3% against MobileNetV3 baselines [30]. For pests at scale, the IP102 dataset of 102 pest categories anchors realistic infestation modelling [31]. The persistent warning across all these sources is field realism: Picon et al. show mobile-captured in-the-wild crop images degrade lab-trained classifiers and propose device-aware training [32], Janarthan et al. build an in-field mobile solution for palm pathology [33], and Picon et al. extend to crop-conditional networks over cell-phone images taken under real field conditions [34]. This work therefore treats lab accuracy as a ceiling signal and field robustness as the binding constraint, which is why evaluation separates controlled classification from cluttered counting.

2.6. Backbones, Compression and Balancing Protocols

The backbone lineage behind Agri-CNX-Edge is itself a literature. Residual learning made very deep networks trainable [35], ConvNeXt modernised pure ConvNets with transformer-inspired design while keeping convolutional efficiency [36], and shifted-window transformers added hierarchical attention with efficient local-global trade-offs [37]. Self-supervision then removed the label bottleneck: vision transformers showed patches plus attention suffice at scale [38], DINO revealed emergent object structure in self-supervised transformers [39], and DINOv2 delivered robust frozen features without supervision [40], culminating in the DINOv3 family used here. Efficiency has a parallel lineage: EfficientNet compounds scaling for accuracy per parameter [41] and MobileNetV3 optimises for phone CPUs through hardware-aware search [42]. Detection contributes its own primitives: the unified real-time YOLO formulation [43] with its community YOLOv5 release [44] and feature pyramids for multi-scale fusion [45], all of which echo in the decoupled anchor-free head and FPN-Lite neck of this work. Adaptation and compression close the loop to the edge: low-rank adaptation inserts tiny trainable matrices into frozen weights [46], deep compression prunes and shares parameters for deployment [47], and integer-arithmetic-only quantization en-

ables on-device inference [48]. For balancing, the survey by Vandenhende et al. systematises multi-task dense prediction trade-offs [49], alongside multi-objective optimisation [50] and gradient surgery for conflicting gradients [51] as heavier alternatives to the homoscedastic weighter adopted here. The conjunction this work tests – one DINOv3 backbone, heterogeneous categorical plus spatial distillation, adaptive weighting and measured phone deployment – appears in none of these sources combined.

Table 1 consolidates studies directly relevant to multi-task or offline mobile agriculture, now spanning more than fifty cited sources across the section. As shown in Table 1, the upper block compares twelve agricultural multi-task systems, the second block compares two offline mobile systems, the third block compares two orchard detection and counting systems, and the lower block places the DINO representation lineage used here. Across the upper block, reported strengths are real, yet narrow: Wen et al. report 69.59% mAP with RMSE 0.44 for dense pest detection and counting, DAM-Net reaches 99.7% on D0 with adaptive loss, Luan et al. report F1 93.74% at 6.25 fps for apple leaf tasks, and EConv-ViT fuses ConvNeXt with Transformer for apple leaves, while Kang et al. provide the closest precedent by joining leaf disease with fruit counting, though only for tomato without pest, grading or DINOv3. The second block shows that offline operation is solved in isolation: PlanteSaine and the MobileNetV3-small edge model run on constrained hardware, with the quantized MobileNetV3-small at 0.93M parameters and 100 fps, yet each remains without unified counting, heterogeneous distillation or multi-task ablation. The third block contributes the spatial machinery this work borrows: Jin et al. harden YOLOv8n for harvesting at 81.68% mAP50 and Hani et al. benchmark yield-mapping methods – both detection plus counting, neither routing detection features into grading. The lower block traces the representation shift from DINOv2-B fusion at 99.67% accuracy without counting or mobile proof to DINOv3 pretraining distilled to ConvNeXt Tiny, which this work adopts as the shared backbone. Read across the Result and Limitation columns, no listed system combines four apple tasks in one DINOv3 backbone with heterogeneous distillation, tested adaptive weighting and measured offline phone deployment. That conjunction is the gap formalised in Section 3.

Table 1: Related work directly relevant to multi-task agriculture, offline mobile deployment and orchard counting. Upper block: agricultural multi-task systems. Second block: offline mobile systems. Third block: orchard detection and counting systems. Lower block: DINO lineage used in this work.

Study	**Year**	**Tasks unified**	**Backbone and edge form**	**Reported result**	**Limitations**
Agricultural multi-task systems					
Wen et al. [10]	2022	Pest detection + counting	YOLOv4 + focal + confluence	69.59% mAP; RMSE 0.44	Pest only; no disease/fruit; no phone measure
Khan et al. [1]	2022	Apple leaf disease	CNN real-time detector	Real-time leaf	Single task; no pest/fruit/counting
Gao et al. [2]	2022	Apple detection + counting	Detector + trunk tracking	Orchard pipeline	Counting only; no classification
DAM-Net [9]	2024	Pest recognition multi-task	Attention + ResNet-50, adapt. loss	99.7% D0; 74.1% IP102	Pest only; no detection; no edge
Kang et al. [4]	2024	Leaf disease + fruit count	Lightweight joint model	Joint inspection	Tomato only; no pest/grading; no DINOv3
Bai et al. [14]	2024	Leaf classific. + severity	DINOv2-B + fusion	99.67%; 90.29% mIoU	No counting; server GPU; mobile future
Sood et al. [8]	2025	Disease + pest	DenseNet-121 + ResNet-50	99.13%; 98.67%	Two-task classifier; no counting; no edge
Luan et al. [3]	2025	Apple disease + segment.	Group pyramid + heads	F1 93.74%; 6.25 fps	Leaf only; no pest/counting
PMJDM [5]	2025	Species + disease detect.	ConvNeXt + dynamic weights	61.83% mAP50; 113 fps	No grading/counting; no phone
CMTL [7]	2025	Boll + pest + stage	Transformer + Jetson Xavier NX	0.913 mAP; 0.832 IoU	Cotton UAV; not apple phone

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Table 1 (continued)

Study	Year	Tasks unified	Backbone and edge form	Reported result	Limitation vs. this work
PlantSDS-Net [6]	2026	Species + disease + severity	Res2Net + MobileNetV2 + attent.	99.89%; 99.58%; 92.18%	Classific. only; no counting/edge
EConv-ViT [15]	2025	Apple leaf disease	ConvNeXt + Transformer + channel attent.	99.2% lab; 79.3% field	Single task; lab-field gap; no edge
Offline mobile systems					
Plante-Saine [11]	2024	Maize/tomato/ onion diagn.	EfficientNetB3 + ANN, TFLite	Offline app	Single-task; no counting/MTL
Schuler et al. [13]	2023	Plant disease edge	MobileNetV3-small ONNX quant.	99.50%; 0.93M; 100 fps	Single task; no MTL/counting
Orchard detection and counting systems					
Jin et al. [23]	2025	Apple detect. + count	Enhanced YOLOv8n + GIoU var.	81.68% mAP50; count 69.39%	Harvesting only; no MTL
Hani et al. [24]	2020	Fruit detect. + yield map	Compared detect. vs tracking	Benchmark of methods	Benchmark only; no system
DINO representation lineage used here					
Simeoni et al. [16]	2026	Foundation pretraining	DINOv3 7B to ConvNeXt Tiny	Frozen SOTA	Pretraining only; no apple system
This work	2026	Leaf + pest + fruit + count	DINOv3-ConvNeXt-Tiny LoRA, ONNX	Sec. 7–9, phone measured	Tested weighting; KD limit documented

3. Research Gap

Section 2 and Table 1 leave a precise conjunction unresolved. Single-task studies leave repeated backbone computation undeployed; multi-task work rarely spans classification *and* spatial counting with distillation for both; and edge claims rarely come with phone

measurements. The gap addressed here is therefore the integration of heterogeneous tasks, hierarchical routing (Eqs. 12–14), categorical-plus-spatial distillation (Eqs. 4–9), adaptive balancing (Eqs. 20–23), and a measured on-device design – with each component ablated rather than assumed. Section 4 implements this integration step-by-step, so every gap item has a corresponding methodological answer.

4. Methodology

This section answers the gap item-by-item in the order it was posed: overview mapping to Fig. 1, teachers supplying Eqs. 1–3, distillation formalised in Eqs. 4–9, backbone and routing in Eqs. 10–14, and heads with optimisation in Eqs. 15–23. Nothing that follows is incidental; each subsection exists because the previous figure or equation requires it.

4.1. System Overview

Agri-CNX-Edge consists of five major components:

1. Dataset curation and preprocessing with cleaning, stratified 70/15/15 splits and task-specific augmentation.
2. Specialized teacher models with ConvNeXt-Small classification teachers fine-tuned from a 120-class PDDD checkpoint and a YOLO26s apple-detection teacher.
3. Shared DINOv3-ConvNeXt-Tiny student backbone with LoRA adapters.
4. FPN-Lite neck and hierarchical feature routing.
5. Task-specific classification heads, a decoupled YOLO detection head, and ROI-Align-based per-apple fruit classification.

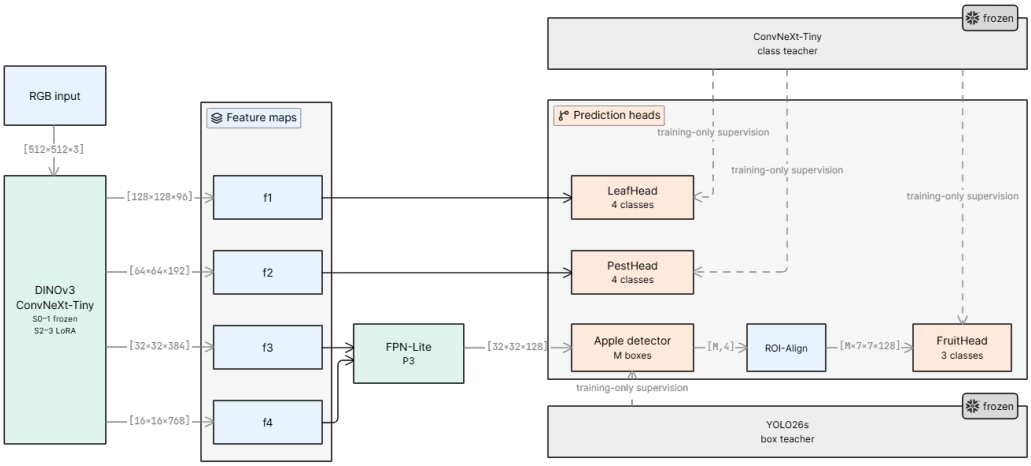


Figure 1: Forward data flow of Agri-CNX-Edge. A shared backbone feeds task-specific heads. Dashed arrows denote training-only teacher supervision detailed in Fig. 3.

As shown in Fig. 1, a single 512-pixel input passes once through the shared DINOv3-ConvNeXt-Tiny backbone, after which the network branches by scale rather than by duplicated encoding. Fine-grained stage 1 features carry texture for leaf classification, mid-scale stage 2 features carry morphology for pest recognition, and the FPN-Lite fusion of stages 3 and 4 carries context for apple detection, from which ROI-Align crops derive per-apple fruit classification. Solid arrows therefore trace inference, which is fully shared up to the heads, while dashed arrows trace training-only teacher supervision formalised in Fig. 3 and Eqs. 4–9. Full stage shapes and routing equalities appear in Eqs. 10–14. This scale-to-task mapping motivates the order of what follows: teachers first, then backbone and routing, then heads and optimisation.

4.2. Teacher Ensemble

Specialized teachers provide task supervision. Classification teachers are ConvNeXt-Small models fine-tuned from a 120-class PDDD checkpoint. The feature dimension is 768 and dropout is 0.5. Training uses focal loss with label smoothing 0.1, class-balanced sampling and heavy augmentation for classes with fewer than 400 training images. It runs up to 100 epochs with an 8-epoch frozen phase, a 5-epoch warm-up and cosine warm restarts. Backbone learning rate is 1×10^{-5}. Head learning rate is 1×10^{-3} in phase 1 and 1×10^{-4} in phase 2.

Three independent classification-teacher seeds are used:

$$S = \{42, 123, 456\}. \tag{1}$$

Eight test-time augmentation views (original, flips, and $\pm 15^\circ/\pm 30^\circ$ rotations) are evaluated per seed. Consequently, the number of teacher inference passes per image for the classification soft labels is:

$$N_{\mathrm{TTA}} = 3 \times 8 = 24. \tag{2}$$

Per-image soft targets are the TTA mean at temperature 1:

$$P_T = \frac{1}{N_{\mathrm{TTA}}} \sum_{k=1}^{N_{\mathrm{TTA}}} P_T^{(k)}, \quad T = 1. \tag{3}$$

The yield task uses a separately trained YOLO26s teacher. Input size is 640 pixels with CIELAB enhancement. Confidence threshold is 0.20, NMS IoU is 0.45 and the limit is 300 detections per image. Normalized boxes and scores are stored as soft labels.

4.3. Categorical and Detection Distillation

For classification, the TTA-averaged teacher probabilities at temperature $T = 1$ are used directly with no further temperature scaling. The student matches them through KL divergence:

$$L_{\mathrm{KL}} = D_{\mathrm{KL}}(P_T \parallel P_S), \tag{4}$$

where P_T and P_S denote teacher and student probability vectors, averaged over samples whose label differs from the ignore index -1. Hard labels are supervised with focal loss at $\gamma = 2.0$, and the two terms are blended with equal weight:

$$L_{\text{cls}} = \alpha L_{\text{focal}} + (1 - \alpha) L_{\text{KL}}, \quad \alpha = 0.5. \tag{5}$$

Whether this blend helps is tested by the no-distillation ablation, not assumed.

For detection, supervision is box-level rather than heatmap-based. Each ground-truth or teacher box with centre (c_x, c_y) is assigned to its grid cell:

$$(i, j) = (\lfloor c_x W \rfloor, \lfloor c_y H \rfloor), \quad W = H = 32. \tag{6}$$

At positive cells, box regression uses Complete IoU loss while objectness and single-class scores use binary cross-entropy:

$$L_{\text{det}} = \lambda_{\text{box}} L_{\text{CIoU}} + \lambda_{\text{obj}} L_{\text{obj}} + \lambda_{\text{cls}} L_{\text{cls}}, \tag{7}$$

with:

$$\lambda_{\text{box}} = 7.5, \quad \lambda_{\text{obj}} = 1.0, \quad \lambda_{\text{cls}} = 0.5. \tag{8}$$

The loss is computed once against ground-truth boxes and once against YOLO teacher box coordinates, with the teacher branch scaled by:

$$L_{\text{yield}} = L_{\text{det}}^{\text{gt}} + \beta L_{\text{det}}^{\text{teacher}}, \quad \beta = 0.5. \tag{9}$$

The fixed β is a stated boundary of the study.

4.4. Student Backbone and Routing

The student backbone is DINOv3-ConvNeXt-Tiny via `timm` (`features_only=True`). For a 512×512 input it yields:

$$f1{:}[B, 96, 128, 128], \quad f2{:}[B, 192, 64, 64], \quad f3{:}[B, 384, 32, 32], \quad f4{:}[B, 768, 16, 16]. \tag{10}$$

Stages 0–1 are fully frozen. Stages 2–3 carry custom LoRA adapters on `mlp.fc1`/`mlp.fc2` with:

$$\text{rank} = 8, \quad \alpha = 16, \quad \text{dropout} = 0.05, \tag{11}$$

all other backbone parameters frozen, leaving only the LoRA A/B matrices trainable. The hierarchical routing is:

$$f1 \rightarrow \text{Leaf}, \quad f2 \rightarrow \text{Pest}, \tag{12}$$

while $f3$ and $f4$ are fused by the FPN-Lite neck into $P3$:

$$P3 = \text{FPN-Lite}(f3, f4) \in \mathbb{R}^{B \times 128 \times 32 \times 32}. \tag{13}$$

The detection and fruit routing from the fused scale is:

$$P3 \rightarrow \text{Apple Detection}, \quad \text{ROIAlign}(P3) \rightarrow \text{Fruit Classification}. \tag{14}$$

Leaf, pest, and fruit heads predict 4, 4, and 3 classes respectively, as configured.

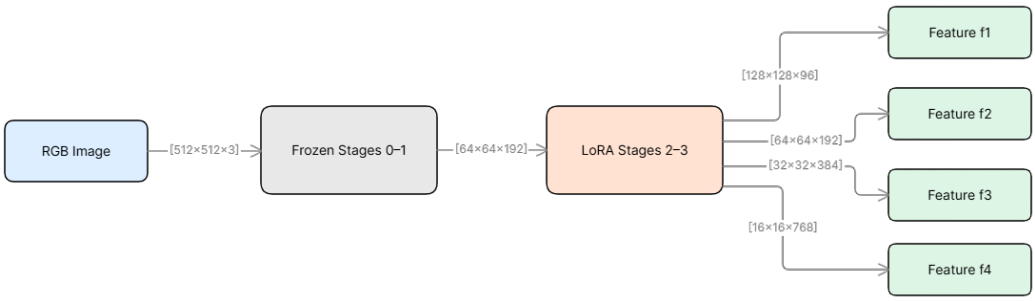


Figure 2: Hierarchical routing from backbone stages to tasks. Details are defined in Section 4.

As shown in Fig. 2, resolution is traded for semantics along the backbone, and each task taps the scale that matches its visual cue. Stage 1 retains high-resolution texture essential for leaf lesions, stage 2 retains mid-resolution shape essential for pest morphology, and the FPN-Lite fusion in Eq. 13 aggregates deep context essential for localising apples under occlusion. Per-apple fruit classification then crops this fused map with ROI-Align, using ground-truth boxes in training and decoded boxes in inference, so grading inherits detection geometry rather than re-encoding the image. The figure thus justifies Eqs. 10–14 and explains why a single scale cannot serve texture, morphology and localisation at once.

4.5. Heads, Detection Decoding, and Optimization

Classification heads share an MBConv design (1×1 reduction to a 256-channel bottleneck, MBConv block with expansion 4 and SE ratio 0.25, global pooling, dropout p=0.3, linear classifier) consuming:

$$\text{Leaf:}f1\ (96\text{ch}) \to 4, \quad \text{Pest:}f2\ (192\text{ch}) \to 4, \quad \text{Fruit:ROI}\ (128\text{ch}, 7 \times 7) \to 3. \tag{15}$$

Detection uses a decoupled anchor-free head on $P3$ emitting:

$$\hat{D} \in \mathbb{R}^{B \times 6 \times 32 \times 32}, \tag{16}$$

where the six channels are $[c_x, c_y, w, h, \text{obj}, \text{cls}]$ with sigmoid applied at loss and post-processing time. Decoding keeps cells with:

$$\text{obj} \times \text{cls} > \text{conf}_{\text{thresh}}, \quad \text{conf}_{\text{thresh}} = 0.3, \tag{17}$$

followed by non-maximum suppression with:

$$\text{nms}_{\text{iou}} = 0.45. \tag{18}$$

Per-image counts are decoded boxes. Task balancing uses homoscedastic uncertainty weighting. For task t the weighter maintains:

$$s_t = \log(\sigma_t^2), \quad \text{initialised to 0.} \tag{19}$$

The multi-task objective is:

$$L_{\mathrm{MTL}} = \sum_{t \in \mathcal{T}} \left(e^{-s_t} L_t + \frac{1}{2} s_t \right), \tag{20}$$

where:

$$\mathcal{T} = \{\mathrm{leaf}, \mathrm{pest}, \mathrm{fruit}, \mathrm{yield}\}. \tag{21}$$

No hard clamping is imposed on s_t; reported weights are precisions:

$$w_t = e^{-s_t}. \tag{22}$$

Distillation needs no extra global coefficients because it is contained inside per-task losses (α in Eq. 5, β in Eq. 9). Hence:

$$L_{\mathrm{total}} = L_{\mathrm{MTL}}. \tag{23}$$

Training is two-phase with 5 epochs for heads and neck with LoRA frozen, then LoRA at 10^{-5} against heads at 10^{-4} with AdamW, cosine schedule, accumulation $\times 4$, clip 1.0, patience 20 and AMP off; the static-weight ablation replaces the weighter with a plain sum.

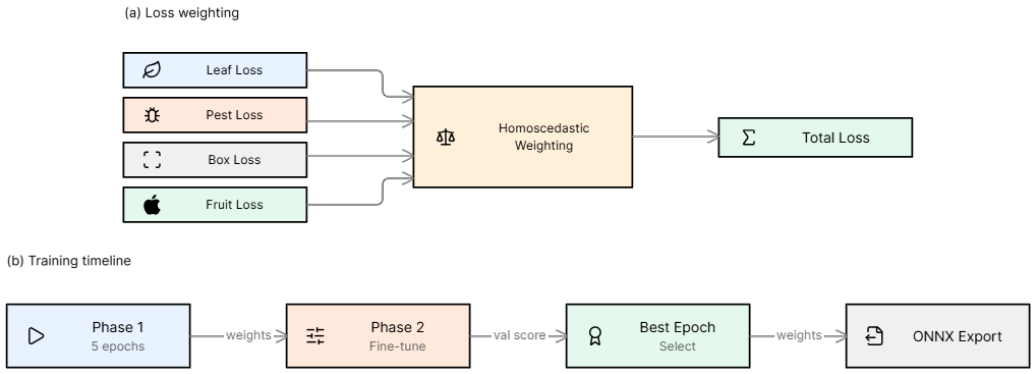


Figure 3: Heterogeneous distillation and two-phase training. Panel a shows classification and detection supervision. Panel b shows the training timeline with best-validation epochs and ONNX export.

As shown in Fig. 3, panel a separates the two teacher streams that supervise one student: TTA-averaged soft targets distil category knowledge into the leaf, pest and fruit heads through Eqs. 4–5, while YOLO boxes distil spatial knowledge into the detection grid through Eqs. 6–9. Data thus flow forward through routing while gradients flow backward from heterogeneous teachers. Panel b places this supervision on a two-phase timeline, with head-only warm-up before joint LoRA adaptation, best validation at epochs 186 and 194, and ONNX export closing training. The figure therefore links Fig. 1 to Eqs. 1–23 and hands over to data and measurement next.

Table 2: Teacher-side dataset cleaning (raw → clean).

Class	Raw	Clean
Leaf: Apple_Mosaic	631	622
Leaf: Apple___Black_rot	621	579
Leaf: Alternaria	631	621
Leaf: Healthy	2276	2152
Pest: xylotrechus	319	296
Pest: aphids	314	270
Pest: leafhoppers	573	567
Pest: spider_mite	299	271
Fruit: Anthracnose	295	275
Fruit: Black Rot	542	524
Fruit: Healthy	1424	1333

5. Dataset Curation and Preprocessing

The methodology requires clean, stratified inputs before any loss in Eqs. 4–23 can be trusted. This section therefore quantifies what the teachers in Section 4 actually saw, so that ceiling effects reported in Section 7 are traceable to data rather than speculation.

Leaf images come from a public PlantVillage mirror on Kaggle [52], of which only the apple leaf classes relevant to the four-class leaf task are used. Fruit images combine a Kaggle apple fruit disease collection [53] with a Mendeley apple fruit dataset, version 6 [54], restricted to the Anthracnose, Black Rot and Healthy classes of Table 2. Pest images come from a Kaggle apple leaf and fruit disease segmentation collection [55], restricted to the four pest classes evaluated here. Apple detection images come from a Kaggle apple quality collection [56]. No repository is used in full; each contributes only its project-relevant subset, and the Raw column of Table 2 reports post-subset, pre-cleaning totals for the classification tasks.

All classification images pass through automated cleaning: images smaller than 64 px or with pixel standard deviation below 5.0 are removed, with perceptual-hash deduplication at threshold 8. As shown in Table 2, retained counts per class remain sufficient for stratified learning. The largest absolute drops are Leaf Healthy from 2276 to 2152 and Fruit Healthy from 1424 to 1333 after duplicate removal, while minority classes such as Pest aphids from 314 to 270 and Fruit Anthracnose from 295 to 275 lose only tens of images. Cleaning is therefore mild and balanced rather than destructive, so the stratified sampling that follows preserves class ratios instead of amplifying imbalance.

Each classification task uses stratified 70/15/15 train/val/test splits with seed 42; yield

detection uses its own split of 1093 train and 264 val. Inputs are 224×224 for classification teachers, 640×640 for the YOLO26s teacher, and 512×512 for the student. Two boundary facts matter later: leaf and pest validation sit near ceiling because teachers are saturated, and no Healthy fruit ROI survives into validation or test.

Segmentation and box labels for apple fruits did not pre-exist and were produced semi-automatically: candidate masks were generated with the base Segment Anything Model from Meta [57] inside X-AnyLabeling [58], operated manually, for apple fruits only. The team then manually verified every image before use, so no SAM proposal enters training or evaluation unverified. Classification labels for leaf, pest and fruit were used as provided by their sources.

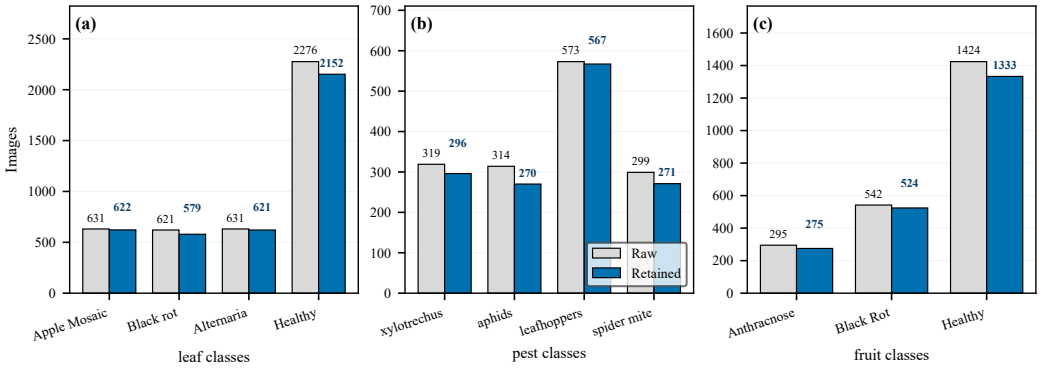


Figure 4: Dataset curation for leaf, pest and fruit teachers with stratified splits. Yield uses its own train and validation split.

As shown in Fig. 4, the curation pipeline applies the same cleaning, deduplication and stratified splitting to leaf, pest and fruit teachers, while yield follows a separate detection split. Corrupt and near-duplicate images are removed without collapsing any minority class, so the class histograms before and after cleaning retain their shape. The distribution is therefore preserved for the class-balanced sampling and heavy augmentation of rare classes described in Section 4, which explains why later ceiling effects can be attributed to task difficulty rather than to data loss.

6. Experimental Setup

With data (Section 5) and model (Section 4) fixed, this section fixes how measurement is done: implementation, optimisation hyperparameters, and metric definitions in Eqs. 24–29. Every number in Section 7 is produced under exactly this protocol, so no result appears without a preceding definition.

The framework is implemented in PyTorch with `timm` backbones, Albumentations and scikit-learn, with ONNX export at opset 17 of leaf logits, pest logits, and the raw detection grid; fruit runs client-side from ROI-Align crops. Students train up to 200 epochs with batch 8 and accumulation $\times 4$ for effective 32 on RTX A4000-class GPU, with AMP off

since CIoU and KL terms proved unstable in FP16, with AdamW at weight decay 10^{-4}, LoRA 10^{-5} and heads, neck and weighter 10^{-4}, cosine annealing to η_{min}=10^{-7}, clipping at norm 1.0, and early stopping with patience 20 and min-delta 10^{-4} on total validation loss.

Classification accuracy is:

$$\text{Accuracy} = \frac{N_{\text{correct}}}{N_{\text{total}}}. \tag{24}$$

Class-wise precision, recall, and F1-score are:

$$\text{Precision} = \frac{TP}{TP + FP}, \tag{25}$$

$$\text{Recall} = \frac{TP}{TP + FN}, \tag{26}$$

$$F1 = 2\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{27}$$

For fruit counting over M images with predicted $\hat{N}_j$ and ground-truth N_j counts:

$$\text{MAE} = \frac{1}{M}\sum_{j=1}^{M}|\hat{N}_j - N_j|, \tag{28}$$

$$\text{RMSE} = \sqrt{\frac{1}{M}\sum_{j=1}^{M}(\hat{N}_j - N_j)^2}. \tag{29}$$

Counting uses decoded boxes at the operating point defined here, namely confidence 0.3 and NMS IoU 0.45. Unless stated otherwise, all detection results use this operating point. All runs use a single seed and early stopping with patience 20. Best epochs are selected by total validation loss.

7. Results

Results are presented in the order the methodology promised: overall reference, frozen control, convergence check, weight inspection, ablation verdict, per-class diagnosis, counting diagnosis, and on-device proof. Each subsection cites its table and figure pair explicitly, and each closes by stating what the next subsection must explain, so the chapter reads as one argument rather than isolated snapshots.

7.1. Overall Performance

The proposed Agri-CNX-Edge model is the reference that later ablations take apart. As shown in Table 3, best-validation accuracy reaches 99.67% for leaf, 100.00% for pest and 96.30% for fruit, with counting MAE 3.5675 and RMSE 6.3767 on 400 yield images. The table thus separates two regimes: leaf and pest sit at ceiling where further gains

Table 3: Best-validation performance of the proposed model.

Task	Metric	Performance
Leaf disease classification	Accuracy (val)	99.67%
Pest recognition	Accuracy (val)	100.00%
Fruit classification	Accuracy (val)	96.30%
Fruit localization/counting	Count MAE	3.5675
Fruit localization/counting	Count RMSE	6.3767

are numerically impossible, while fruit accuracy trails by nearly four points and counting errors remain well above zero. As shown in Fig. 5, the accuracy panel makes this split visual by placing leaf and pest bars at the top of the scale against a distinctly lower fruit bar, while the error panel places MAE and RMSE on a count scale that exposes the spatial task as the remaining challenge. Together, table values and figure shapes establish where diagnosis must focus. Leaf and pest therefore offer no further signal, so the next subsection turns to fruit and counting.

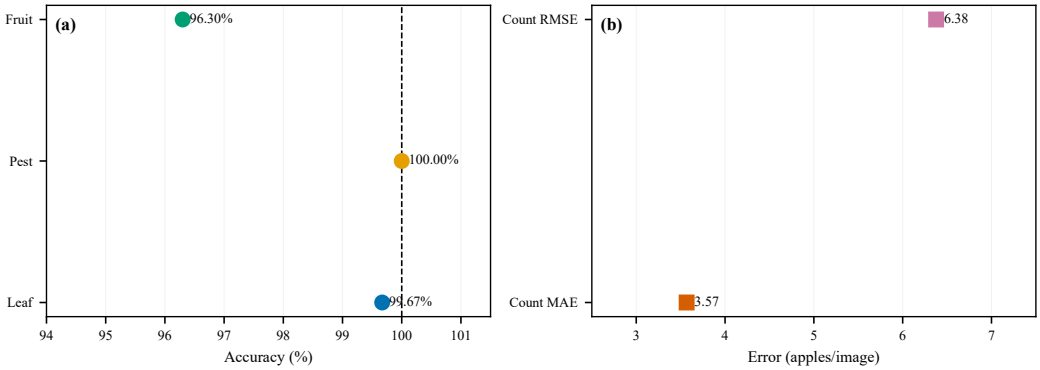


Figure 5: Best-validation snapshot of the proposed model. Accuracies and counting errors on validation.

7.2. Frozen Versus Proposed Configuration

Checkpoint inspection confirms attribution, as LoRA adapters are present only in the proposed checkpoint. As shown in Table 4, adaptation improves leaf marginally from 99.50% to 99.67%, a gain of 0.17 points, and holds pest equal at 100.00%, but reduces fruit accuracy from 98.42% to 96.30%, a loss of 2.12 points, while increasing counting error from MAE 3.4300 to 3.5675 and RMSE 6.0017 to 6.3767. The table therefore documents an inversion: the only task that improves is already at ceiling, while the headroom tasks of fruit grading and dense localisation degrade. As shown in Fig. 6, paired deltas visualise

Table 4: Frozen and proposed configurations with paired deltas.

Metric	Frozen	Proposed	Change
Leaf Accuracy	99.50%	99.67%	+0.17 pp
Pest Accuracy	100.00%	100.00%	0.00 pp
Fruit Accuracy	98.42%	96.30%	-2.12 pp
Count MAE	3.4300	3.5675	+0.1375
Count RMSE	6.0017	6.3767	+0.3750

the same inversion, with a near-zero leaf bar against clearly negative fruit and counting bars. The pattern is reported as observed rather than rationalised, and it motivates the full ablation that isolates adaptation, distillation, weighting and warm-up.

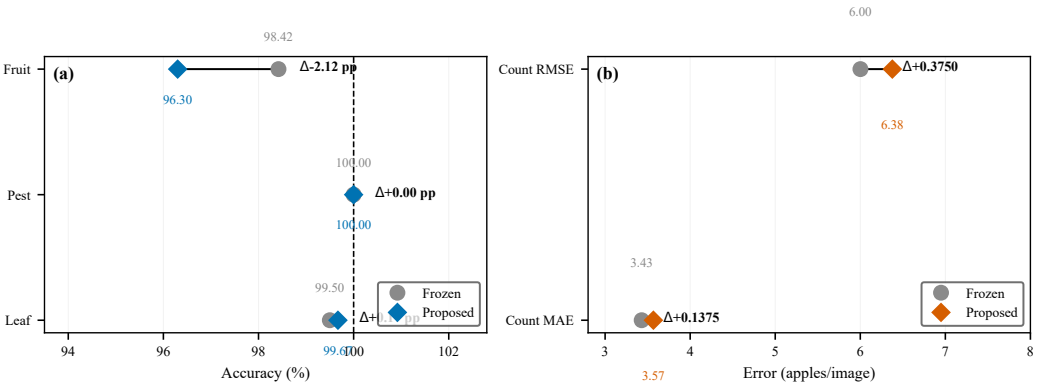


Figure 6: Frozen and proposed configurations with paired deltas.

7.3. Training Convergence

Convergence timing does not explain the fruit gap. As shown in Table 5, best epochs are 186 for the proposed run and 194 for the frozen run, with best validation losses of -1.9251 and -1.9271 and training losses of -0.7437 and -0.6637 at those epochs. Both the timing and the loss levels are near identical, so neither under-training nor overfitting distinguishes the two runs. As shown in Fig. 7, panel a shows overlapping training and validation loss trajectories across 200 epochs, while panel b shows per-task validation accuracy in which leaf and pest flatten early near 100% and fruit oscillates most across epochs. The visual message matches the tabular one: optimisation length is not the bottleneck. The cause must therefore lie in task weighting rather than training length, which leads to the next subsection.

Table 5: Training convergence at best-validation epochs.

Training Statistic	**Proposed**	**Frozen**
Best epoch	186	194
Best validation loss	-1.9251	-1.9271
Training loss at best epoch	-0.7437	-0.6637

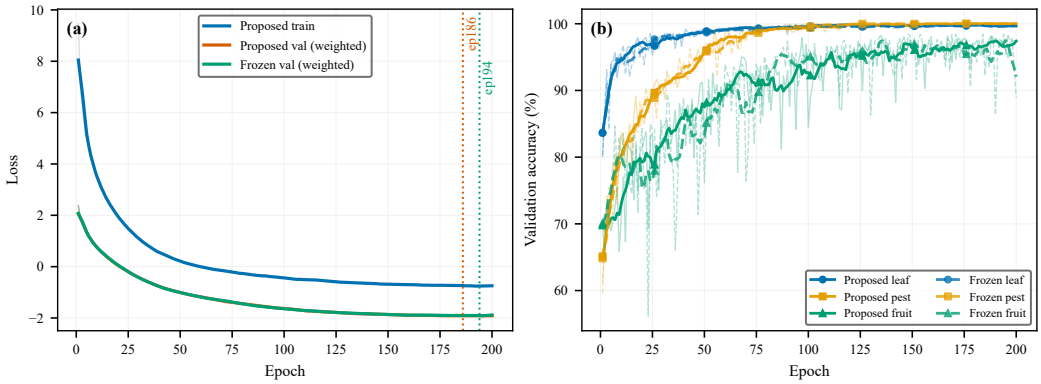


Figure 7: Training dynamics over 200 epochs. Panel a shows training and validation loss. Panel b shows per-task validation accuracy.

7.4. Learned Task Weights

Learned precisions in Eq. 22 are optimisation coefficients rather than measures of importance. As shown in Tables 6 and 7 and Fig. 8, fruit receives the largest weight in both runs, 18.8170 for the proposed run at epoch 186 and 18.4710 for the frozen run at epoch 194, followed by leaf at 6.9591 and 7.1471, pest at 5.7344 and 5.7203, and yield at 0.1780 and 0.1760, with nearly identical ordering across runs. The figure makes this hierarchy immediate by placing the fruit bar far above the rest in both runs, while the tables supply the exact precisions. Such stability across adaptation states shows that the weighting reflects loss scale rather than task priority. Whether this learned balance matters is tested by replacing it with equal weights in the ablation.

Table 6: Terminal task precisions, proposed (epoch 186).

Task	Terminal Weight
Fruit	18.8170
Leaf	6.9591
Pest	5.7344
Yield	0.1780

Table 7: Terminal task precisions, frozen (epoch 194).

Task	Terminal Weight
Fruit	18.4710
Leaf	7.1471
Pest	5.7203
Yield	0.1760

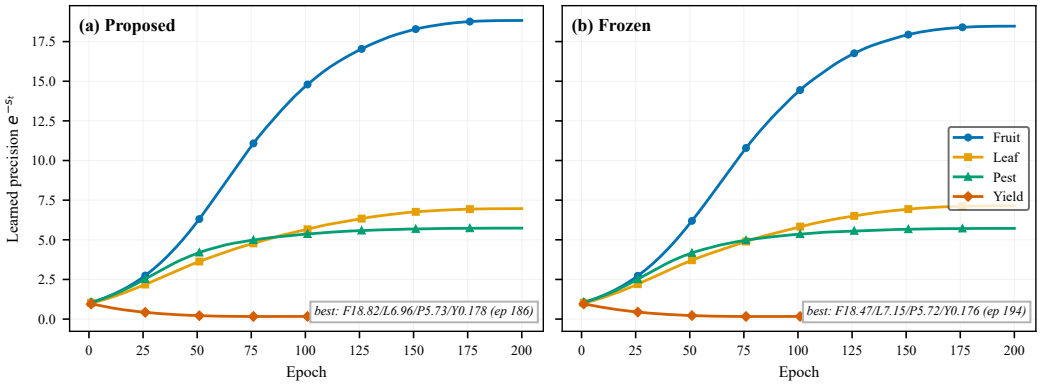


Figure 8: Learned task precisions for proposed and frozen runs.

8. Ablation Study

Each configuration removes one component while holding the rest fixed. As shown in Table 8, which reports best-validation epochs, removing distillation without KD matches or improves every validation metric, with leaf rising from 99.67% to 100.00%, pest holding at 100.00%, fruit rising from 96.30% to 98.47%, MAE falling from 3.5675 to 3.0175 and RMSE falling from 6.3767 to 5.2714, while replacing learned weighting with equal weights collapses validation fruit to 81.43%. On held-out test the same ranking holds and widens: without KD reaches 99.14% fruit with MAE 3.08 against proposed 95.69% with MAE 3.53, while static weighting falls further to 70.26% fruit. Adaptation without LoRA at 94.69% and without warm-up at 97.18% remain within run-to-run noise. As shown in Fig. 9, the radar makes the ranking visual by enclosing the largest area for the no-distillation run and the smallest for static weighting, with the exact values tabulated alongside. The static-weighting run stopped after 103 epochs and is therefore flagged as incomplete, which Section 13 revisits.

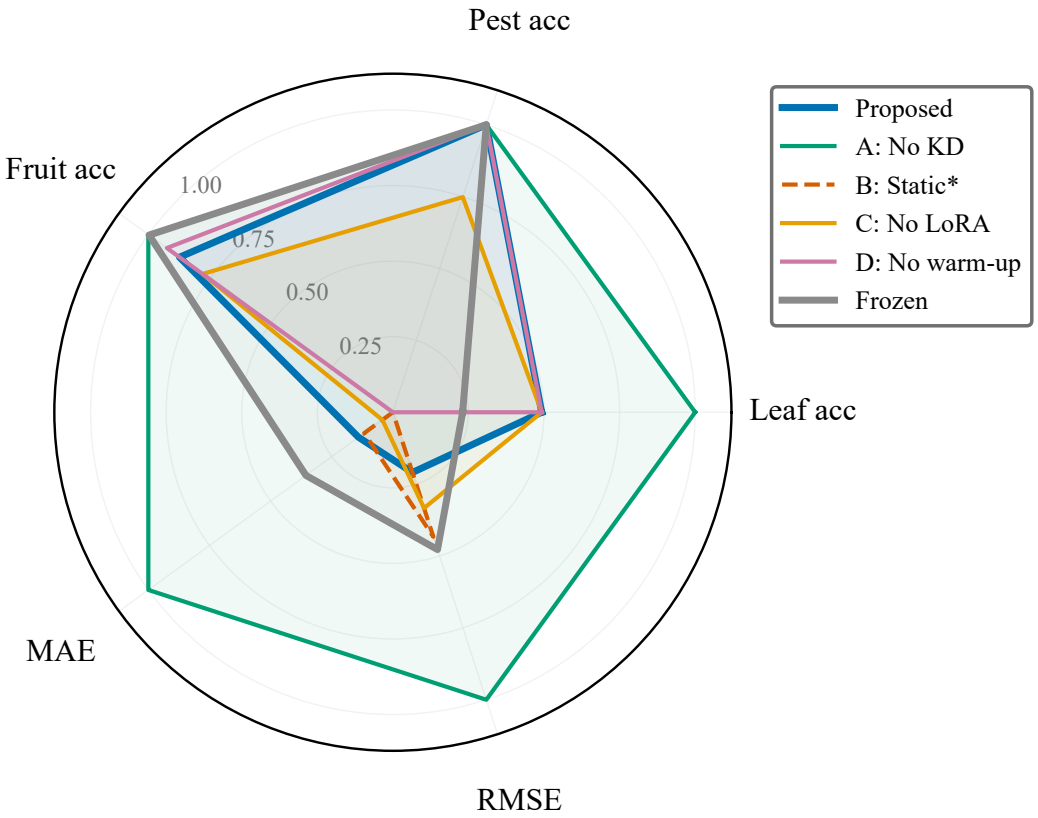


Figure 9: Component ablation. Radar shows relative ranking and the table in the figure lists exact values.

Table 8: Component ablation at best-validation epochs. Static weighting is incomplete and is interpreted only as collapse evidence.

Config	Leaf	Pest	Fruit	MAE	RMSE
Proposed	99.67%	100.00%	96.30%	3.5675	6.3767
No KD	100.00%	100.00%	98.47%	3.0175	5.2714
Static	99.35%	98.33%	81.43%	3.5825	6.0640
No LoRA	99.67%	99.58%	94.69%	3.6300	6.2032
No warm-up	99.67%	100.00%	97.18%	3.6575	6.6695
Frozen	99.50%	100.00%	98.42%	3.4300	6.0017

9. Class-wise Classification Analysis

The following subsections move from validation to held-out test. Leaf and pest confirm ceiling behaviour while fruit reveals headroom. Each pair of figure and table tells one story, with the figure showing error location and the table quantifying it.

9.1. Leaf Disease Classification

Held-out predictions on 623 images give a clean sweep. The evaluation uses the latest checkpoint because only it was present on the test machine, as noted in Section 13. As shown in Fig. 10 and Table 9, the confusion matrix is perfectly diagonal with zero off-diagonal entries, and all four classes reach precision, recall and F1 of 1.0000. Such perfection across Apple Mosaic, Black rot, Alternaria and Healthy confirms that texture cues from stage 1 are fully sufficient. Leaf is therefore at ceiling, consistent with the 99.67% validation accuracy, and needs no further diagnosis.

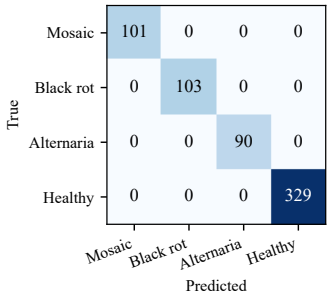


Figure 10: Leaf confusion matrix on held-out test.

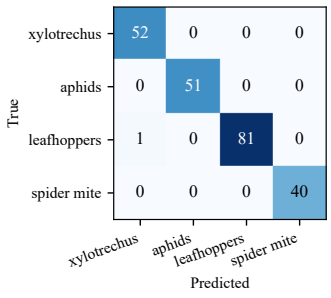


Figure 11: Pest confusion matrix on held-out test.

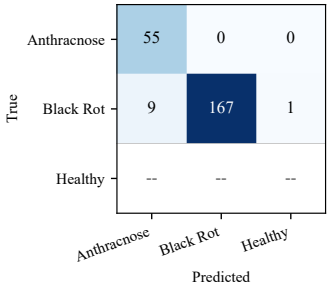


Figure 12: Fruit confusion matrix on ROI test.

Table 9: Leaf test results by class.

Class	Precision	Recall	F1
Apple_Mosaic	1.0000	1.0000	1.0000
Apple___Black_rot	1.0000	1.0000	1.0000
Alternaria	1.0000	1.0000	1.0000
Healthy	1.0000	1.0000	1.0000

9.2. Pest Recognition

One error occurs in 225 test images. As shown in Fig. 11 and Table 10, a single leafhoppers image is predicted as xylotrechus, which lowers xylotrechus precision to 0.9811 and leafhoppers recall to 0.9878, with corresponding F1 of 0.9905 and 0.9939, while aphids and spider mite remain perfect at 1.0000. The confusion matrix is therefore diagonal except for one cell, a pattern characteristic of morphological overlap between small insects rather than systematic failure. Pest is thus also at ceiling at 99.56% test accuracy, matching validation.

Table 10: Pest test results by class.

Class	Precision	Recall	F1
xylotrechus	0.9811	1.0000	0.9905
aphids	1.0000	1.0000	1.0000
leafhoppers	1.0000	0.9878	0.9939
spider_mite	1.0000	1.0000	1.0000

9.3. Fruit Classification

Fruit is evaluated as a head diagnostic on ground-truth ROIs. No Healthy ROI exists in validation or test, so generalization for that class remains untested. As shown in Fig. 12 and Table 11, Anthracnose reaches recall 1.0000 but precision only 0.8594 with F1 0.9244, while Black Rot reaches precision 1.0000 but recall only 0.9435 with F1 0.9709, so confusion concentrates between Black Rot and Anthracnose in one direction. This asymmetry defines fruit at 95.69% test accuracy as the headroom task. It links back to the validation gap of 96.30% and forward to counting, which shares the same fused features.

Table 11: Fruit ROI test results by class. Healthy has no test ROIs.

Class	Precision	Recall	F1
Anthracnose	0.8594	1.0000	0.9244
Black Rot	1.0000	0.9435	0.9709
Healthy	—	—	—

10. Fruit Counting and Spatial Evaluation

Counting correlates well but undercounts systematically on dense clusters. As shown in Table 12, on 400 yield images MAE is 3.5300 and RMSE 6.1323 with Pearson r=0.8508 and R^2=0.5700, while mean signed error is −3.1000. High correlation with negative bias thus separates ranking quality from systematic undercounting: the model orders images correctly yet predicts too few apples per image. As shown in Fig. 13, predictions cluster below the identity line with widening spread at high counts, so sparse orchard images lie near the line while dense clusters fall clearly beneath it, visualising the same bias numerically reported in the table.

As shown in Fig. 14, sparse cases with one to five apples match ground truth exactly, while dense clusters with overlapping canopies miss occluded instances, which is the spatial counterpart of the −3.1 bias in Table 12. As shown in Fig. 15, detector confidence maps are sharp on isolated apples and decay in clutter, so missed detections coincide with low confidence regions rather than with random error. These figures complete the spatial half of the claim opened in Fig. 1 and point to threshold calibration for dense scenes as the natural remedy.

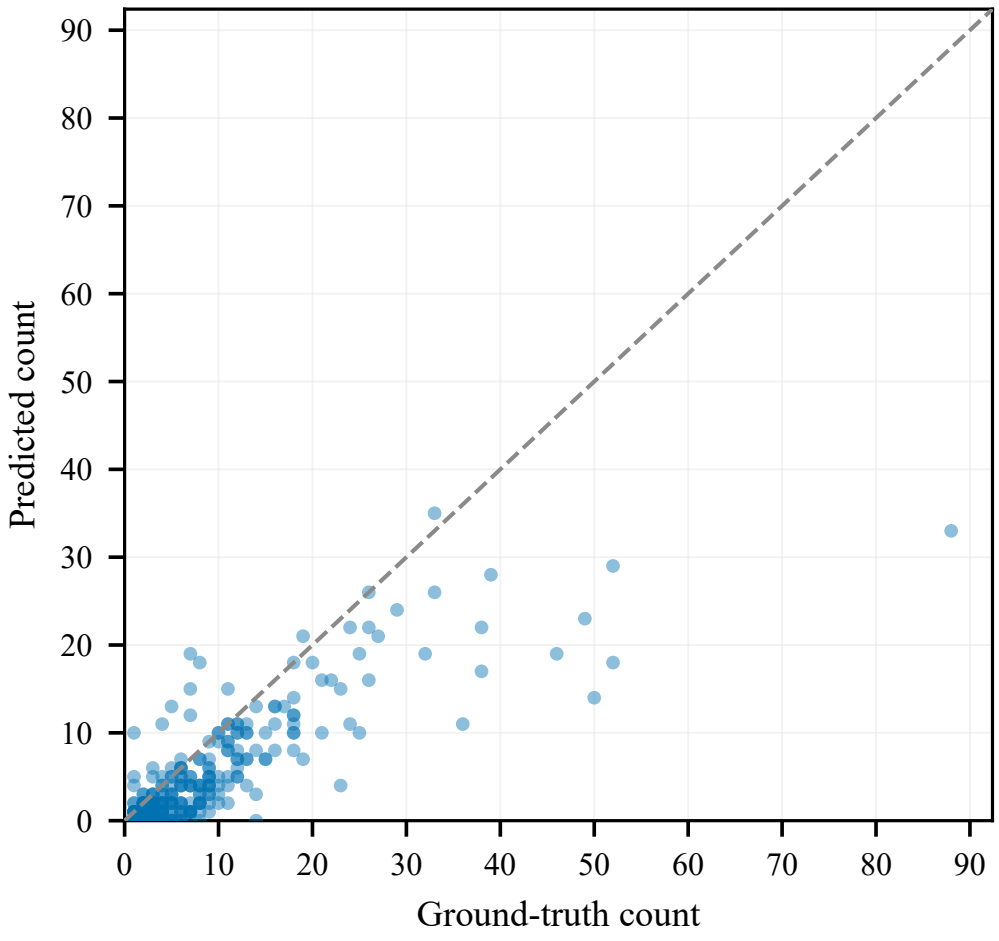


Figure 13: Predicted against ground-truth counts with identity reference.



Figure 14: Ground-truth and predicted boxes from sparse to dense cases with per-image counts.

Table 12: Counting performance on 400 yield images.

Metric	Value
Count MAE	3.5300
Count RMSE	6.1323
Pearson correlation	0.8508
R^2	0.5700
Mean signed error	-3.1000

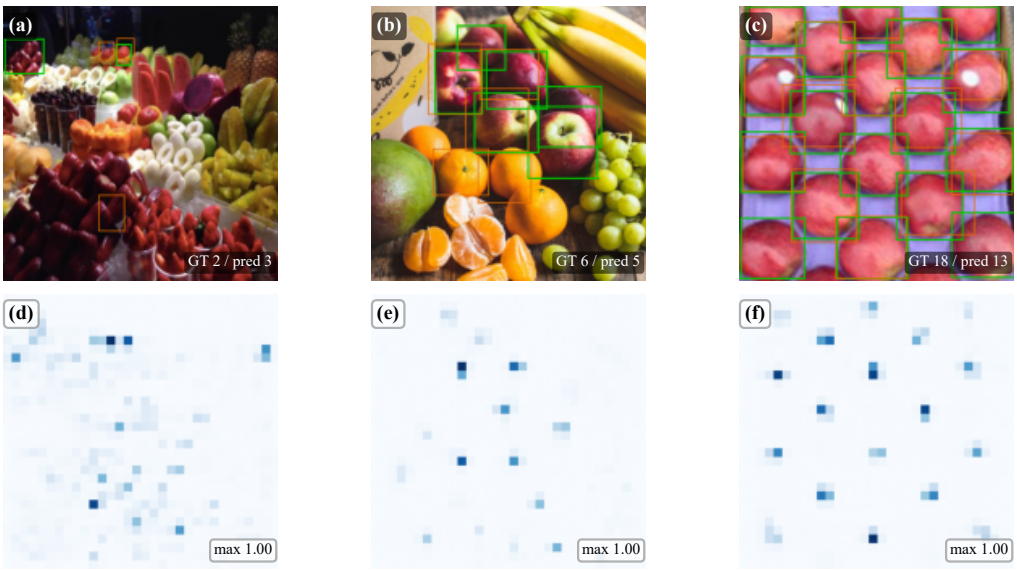


Figure 15: Inputs with counts and corresponding detector confidence maps.

11. Edge Deployment Analysis

Shared computation must be measured rather than asserted. Tests use a Samsung A17 5G with ONNX Runtime under a uniform protocol with an accuracy pass over 30 yield and 20 pest images and a latency pass over 100 measured plus 10 warm-up inferences per mode, using the same app, model, images and code in both provider modes. Full protocol details are in the accompanying benchmark description. As shown in Table 13, the unified 120.7 MB student with 28.28 M parameters including 0.46 M LoRA is substantially smaller than the teachers at 188.8, 188.8 and 19.4 MB, phone yield MAE is 2.9 for both CPU and NNAPI over 30 images with 20 of 20 pest agreement to laptop in both modes and zero CPU to NNAPI prediction differences. Mean latency is 5749 ms for CPU and 5604 ms for NNAPI, with p50 of 7041 and 6763 ms and p95 near 7381 and 7406 ms, peak PSS of 765 to 807 MB and fully offline operation. As shown in Fig. 16, file-size bars visualise the sharing dividend of one student against three teachers, and latency plots show coincident provider modes, which indicates CPU fallback on this device. Table quantifies and figure visualises the efficiency claim of the conclusion.

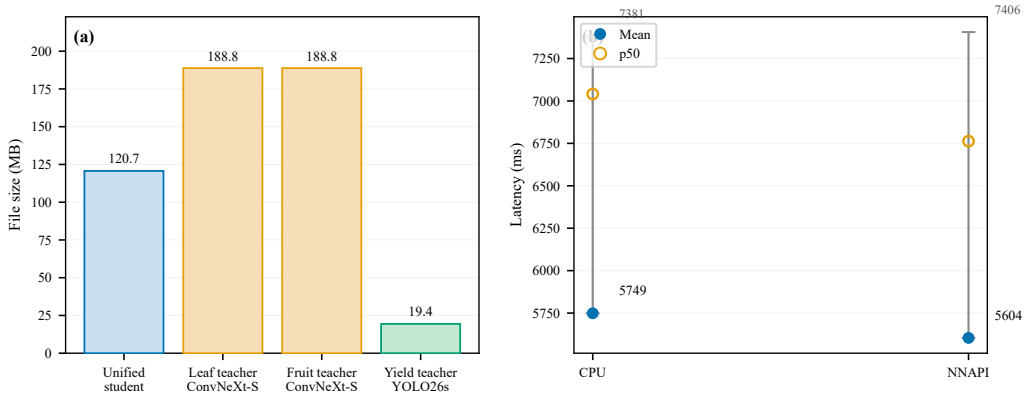


Figure 16: On-device cost. File sizes against teachers and latency by provider mode with identical predictions.

Table 13: On-device results on Samsung A17 5G.

Metric	Measured
Backbone params	28.28 M including 0.46 M LoRA
ONNX file	120.7 MB at opset 17
Torch checkpoint at epoch 186	148.5 MB
Leaf / fruit / YOLO teacher files	188.8 / 188.8 / 19.4 MB
Pest teacher file	absent and excluded
Yield MAE, phone CPU/NNAPI	2.9 / 2.9
Pest agreement, phone–laptop	20/20 both modes
CPU–NNAPI prediction diffs	0, identical
Latency mean (ms)	5749 / 5604
Latency p50 (ms)	7041 / 6763
Latency p95, approx. (ms)	7381 / 7406
Peak PSS	765–807 MB
Battery drain	9% total from 84 to 75% over ∼320 inferences in ∼28 min; 3% per 100-inference latency run
Device power	≈3.7 W device-level from drain on 5000 mAh cell, coarse screen-on measure
Test conditions	saver off, unplugged, min brightness
Offline operation	Yes with on-device ORT

12. Conclusion and Discussion

This section synthesises what Tables 3–13 and Figs. 5–16 jointly imply, rather than repeating them. The held-out test establishes a working unified model with ceiling leaf and pest performance and strong counting correlation. It refuses the expected conclusion, as removing distillation matches or improves every metric and freezing the backbone ties the proposed system. Distillation therefore does not help under saturated teachers and ceiling tasks with fixed strengths and one seed per configuration, while adaptation remains within noise. The positive result is unambiguous, as static weighting collapses fruit accuracy and thus validates adaptive weighting as essential. Learned precisions are optimisation coefficients rather than measures of importance. On device, the surviving contribution is efficiency with identical numerics and no acceleration on the test hardware.

Agri-CNX-Edge unifies leaf, pest, fruit, and counting/detection in one offline-capable DINOv3-ConvNeXt model with test 100.00% leaf, 99.56% pest and 95.69% fruit and counting MAE 3.53 at r=0.85 and R^2=0.57. Against frozen at 95.69% and 3.43 and without KD at 99.14% and 3.08, the measured contribution is the unified efficiency system at

120.7 MB offline with provider-identical numerics plus the validation of adaptive weighting with static weighting at 70.26% – alongside a documented negative distillation result under saturated-teacher conditions. Class-wise, correlation, qualitative, and on-device evidence are measured; confirmatory work is multi-seed repeats, static-weight completion, and quantized builds.

13. Limitations and Future Work

Boundary conditions are stated plainly. The static-weighting run is incomplete and is cited only as collapse evidence until completed. Healthy fruit has no validation or test ROIs, so its generalization is untested. The proposed system was evaluated from the latest checkpoint rather than the best-validation epoch because only it was present at evaluation time. Counting shows systematic undercounting that calls for threshold calibration rather than ranking analysis. Latency tails are approximate, battery readings are coarse and the latency subset is train sourced, while accuracy rests on validation and held-out test. Pest phone images come from the training pool and serve only as agreement sanity. A pilot on an older low-end phone was excluded as impractical, so ultra-low-end deployment requires quantization.

Future work can investigate:

- completing the static-weighting run and repeating configurations over seeds,
- threshold calibration for dense clusters,
- post-training quantization for ultra-low-end deployment,
- Healthy-fruit data collection,
- larger-scale orchard evaluation, and
- comparison with single-task specialist baselines.

Conflict of Interest

The authors declare that there is no conflict of interest regarding the publication of this paper.

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Notes on contributor(s)

All the author(s) contributed to the conception and design of the study. Data were collected by Aarya Balwadkar, Afifa Bintul Hasan, Eccha Bansal and Aparna Nair. Aarya Balwadkar, Afifa Bintul Hasan, Eccha Bansal and Ankur Goyal performed the data analysis. The first draft of the manuscript was written by Aarya Balwadkar, Eccha Bansal and Ankur Goyal reviewed and edited the manuscript. All authors read and approved the final version of the manuscript.

Ethics Approval

Not Applicable.

Data Availability Statement

The datasets used and/or analyzed during the current study are available from any authors from upon request.

Clinical trial

Clinical trial number: not applicable.

Consent to Participate

Not applicable.

Consent to Publish

This study does not include any identifiable human data.

References

[1] S. I. Khan, S. M. Quadri, S. Banday, and J. L. Shah, "Deep diagnosis: A real-time apple leaf disease detection system based on deep learning," *Computers and Electronics in Agriculture*, vol. 198, 2022, Art. no. 107093, doi:10.1016/j.compag.2022.107093.

[2] F. Gao et al., "A novel apple fruit detection and counting methodology based on deep learning and trunk tracking in modern orchard," *Computers and Electronics in Agriculture*, vol. 197, 2022, Art. no. 107000, doi:10.1016/j.compag.2022.107000.

[3] P. Luan et al., "Accurate and efficient recognition of mixed diseases in apple leaves using a multi-task learning approach," *Agriculture*, vol. 16, no. 1, 2025, Art. no. 71, doi:10.3390/agriculture16010071.

[4] R. Kang et al., "Toward real scenery: A lightweight tomato growth inspection algorithm for leaf disease detection and fruit counting," *Plant Phenomics*, vol. 6, 2024, Art. no. 0174, doi:10.34133/plantphenomics.0174.

[5] Y. Li et al., "PMJDM: a multi-task joint detection model for plant disease identification," *Frontiers in Plant Science*, vol. 16, 2025, Art. no. 1599671, doi:10.3389/fpls.2025.1599671.

[6] S. Sharma et al., "A multi-feature fusion deep neural network with attention for multi-task crop monitoring," *Eng. Res. Express*, 2026, in press, doi:10.1088/2631-8695/ae5913.

[7] S. Umirzakova et al., "A unified transformer model for simultaneous cotton boll detection, pest damage segmentation, and phenological stage classification," *Drones*, vol. 9, no. 8, 2025, Art. no. 555, doi:10.3390/drones9080555.

[8] V. Sood et al., "Towards automated crop protection: fusion of DenseNet121-ResNet50 model for disease detection and pest recognition," *Int. J. Syst. Assur. Eng. Manag.*, 2025, doi:10.1007/s13198-025-02975-2.

[9] H. Zhang et al., "Enhanced pest recognition using multi-task deep learning with the discriminative attention multi-network," *Applied Sciences*, vol. 14, no. 13, 2024, Art. no. 5543, doi:10.3390/app14135543.

[10] C. Wen et al., "Pest-YOLO: A model for large-scale multi-class dense and tiny pest detection and counting," *Frontiers in Plant Science*, vol. 13, 2022, Art. no. 973985, doi:10.3389/fpls.2022.973985.

[11] O. Appiah et al., "PlanteSaine: An artificial intelligent empowered mobile application for pests and disease management," *Agriculture*, vol. 14, no. 8, 2024, Art. no. 1252, doi:10.3390/agriculture14081252.

[12] B. Wang et al., "An ultra-lightweight efficient network for image-based plant disease and pest infection detection," *Precision Agriculture*, vol. 24, 2023, pp. 1836–1861, doi:10.1007/s11119-023-10020-0.

[13] J. Schuler et al., "Plant disease detection model for edge computing devices," *Frontiers in Plant Science*, vol. 14, 2023, Art. no. 1308528, doi:10.3389/fpls.2023.1308528.

[14] C. Bai et al., "DINOV2-FCS: a model for fruit leaf disease classification and severity prediction," *Frontiers in Plant Science*, vol. 15, 2024, Art. no. 1475282, doi:10.3389/fpls.2024.1475282.

[15] X. Huang et al., "EConv-ViT: A strongly generalized apple leaf disease classification model based on the fusion of ConvNeXt and Transformer," *Inf. Process. Agric.*, 2025, doi:10.1016/j.inpa.2025.03.001.

[16] O. Siméoni et al., "DINOv3," *Trans. Mach. Learn. Res.*, 2026.

[17] Q. Huang et al., "Knowledge distillation facilitates the lightweight and efficient plant diseases detection model," *Plant Phenomics*, vol. 5, 2023, Art. no. 0062, doi:10.34133/plantphenomics.0062.

[18] A. Kendall, Y. Gal, and R. Cipolla, "Multi-task learning using uncertainty to weigh losses for scene geometry and semantics," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 7482–7491, doi:10.1109/CVPR.2018.00781.

[19] Z. Chen, V. Badrinarayanan, C.-Y. Lee, and A. Rabinovich, "GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks," in *Proceedings of the International Conference on Machine Learning (ICML)*, 2018, pp. 794–803.

[20] B. Liu, X. Liu, X. Jin, P. Stone, and Q. Liu, "Conflict-averse gradient descent for multi-task learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021.

[21] V. J. Reddi et al., "MLPerf mobile inference benchmark: An industry-standard open-source machine learning benchmark for on-device AI," in *Proceedings of Machine Learning and Systems (MLSys)*, vol. 4, 2022, pp. 352–369.

[22] X. Liu et al., "A comprehensive benchmark of deep learning libraries on mobile devices," in *Proceedings of the ACM Web Conference (WWW)*, 2022, doi:10.1145/3485447.3512148.

[23] T. Jin, X. Han, P. Wang, Z. Zhang, J. Guo, and F. Ding, "Enhanced deep learning model for apple detection, localization, and counting in complex orchards for robotic arm-based harvesting," *Smart Agricultural Technology*, vol. 10, 2025, Art. no. 100784, doi:10.1016/j.atech.2025.100784.

[24] N. Hani, P. Roy, and V. Isler, "A comparative study of fruit detection and counting methods for yield mapping in apple orchards," *J. Field Robot.*, vol. 37, no. 2, 2020, pp. 263–282, doi:10.1002/rob.21902.

[25] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," *Frontiers in Plant Science*, vol. 7, 2016, Art. no. 1419, doi:10.3389/fpls.2016.01419.

[26] K. P. Ferentinos, "Deep learning models for plant disease detection and diagnosis," *Computers and Electronics in Agriculture*, vol. 145, 2018, pp. 311–318, doi:10.1016/j.compag.2018.01.009.

[27] V. S. Dhaka et al., "A survey of deep convolutional neural networks applied for prediction of plant leaf diseases," *Sensors*, vol. 21, no. 14, 2021, Art. no. 4749, doi:10.3390/s21144749.

[28] J. Liu and X. Wang, "Plant diseases and pests detection based on deep learning: a review," *Plant Methods*, vol. 17, 2021, Art. no. 22, doi:10.1186/s13007-021-00722-9.

[29] Shoaib et al., "Leveraging deep learning for plant disease and pest detection: a comprehensive review and future directions," *Frontiers in Plant Science*, vol. 16, 2025, Art. no. 1538163, doi:10.3389/fpls.2025.1538163.

[30] S. Banerjee, T. Mallick, A. Chakroborty, H. N. Saha, and N. T. Takur, "Evaluation of state-of-the-art deep learning techniques for plant disease and pest detection," *Comput. Mater. Contin.*, vol. 85, 2025, pp. 121–180, doi:10.32604/cmc.2025.065250.

[31] X. Wu et al., "IP102: A large-scale benchmark dataset for insect pest recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, doi:10.1109/CVPR.2019.00899.

[32] A. Picon, A. Alvarez-Gila, M. Seitz, A. Ortiz-Barredo, J. Echazarra, and A. Johannes, "Deep convolutional neural networks for mobile capture device-based crop disease classification in the wild," *Computers and Electronics in Agriculture*, vol. 161, 2019, pp. 280–290, doi:10.1016/j.compag.2018.04.002.

[33] S. Janarthan et al., "P2OP–Plant Pathology on Palms: A deep learning-based mobile solution for in-field plant disease detection," *Computers and Electronics in Agriculture*, vol. 202, 2022, Art. no. 107371, doi:10.1016/j.compag.2022.107371.

[34] A. Picon et al., "Crop conditional convolutional neural networks for massive multi-crop plant disease classification over cell phone acquired images taken on real field conditions," *Computers and Electronics in Agriculture*, vol. 167, 2019, Art. no. 105093, doi:10.1016/j.compag.2019.105093.

[35] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778, doi:10.1109/CVPR.2016.90.

[36] Z. Liu et al., "A ConvNet for the 2020s," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 11976–11986, doi:10.1109/CVPR52688.2022.01167.

[37] Z. Liu et al., "Swin Transformer: Hierarchical vision transformer using shifted windows," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 10012–10022, doi:10.1109/ICCV48922.2021.00986.

[38] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021.

[39] M. Caron et al., "Emerging properties in self-supervised vision transformers," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 9650–9660, doi:10.1109/ICCV48922.2021.00951.

[40] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," *Trans. Mach. Learn. Res.*, 2024.

[41] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proceedings of the International Conference on Machine Learning (ICML)*, 2019, pp. 6105–6114.

[42] A. Howard et al., "Searching for MobileNetV3," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 1314–1324, doi:10.1109/ICCV.2019.00140.

[43] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788, doi:10.1109/CVPR.2016.91.

[44] G. Jocher et al., "Ultralytics YOLOv5," software release, 2020.

[45] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 2117–2125, doi:10.1109/CVPR.2017.106.

[46] E. J. Hu et al., "LoRA: Low-rank adaptation of large language models," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2022.

[47] S. Han, H. Mao, and W. J. Dally, "Deep compression: Compressing deep neural networks with pruning, trained quantization and Huffman coding," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2016.

[48] B. Jacob et al., "Quantization and training of neural networks for efficient integer-arithmetic-only inference," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, doi:10.1109/CVPR.2018.00286.

[49] S. Vandenhende, S. Georgoulis, W. Van Gansbeke, M. Proesmans, D. Dai, and L. Van Gool, "Multi-task learning for dense prediction tasks: A survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 7, 2022, pp. 3614–3633, doi:10.1109/TPAMI.2021.3054719.

[50] O. Sener and V. Koltun, "Multi-task learning as multi-objective optimization," in *Adv. Neural Inf. Process. Syst.*, vol. 31, 2018.

[51] T. Yu et al., "Gradient surgery for multi-task learning," in *Adv. Neural Inf. Process. Syst.*, vol. 33, 2020.

[52] "PlantVillage dataset (mirror)," Kaggle. `https://www.kaggle.com/datasets/abdallahalidev/plantvillage-dataset`.

[53] "Apple fruit disease images dataset," Kaggle. `https://www.kaggle.com/datasets/anilsandhii/apple-fruit-disease-images-dataset`.

[54] "Apple fruit dataset," Mendeley Data, v6. https://data.mendeley.com/datasets/9zgkwwv9j8/6, doi:10.17632/9zgkwwv9j8.6.

[55] "Apple leaf and fruit disease segmentation dataset," Kaggle. https://www.kaggle.com/datasets/yaldanz/apple-leaf-and-fruit-disease-segmenattion.

[56] "Apple quality dataset," Kaggle. https://www.kaggle.com/datasets/snehamahata/apple-quality-dataset.

[57] A. Kirillov et al., "Segment Anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, doi:10.1109/ICCV51070.2023.00371.

[58] W. Wang, "Advanced auto labeling solution with added features," GitHub repository, CVHub, 2023. https://github.com/CVHub520/X-AnyLabeling.